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Introduction to Microgrids

A **microgrid** is a small, local electrical power system that can generate, store, control, and deliver electricity to a specific area. This area could be a university campus, hospital, factory, remote community, military base, data center, or even a small neighborhood. In simple words, a microgrid is like a smaller version of the main power grid, but it is designed to serve a local area and can be controlled more directly.

Structure/Components

A microgrid usually contains several important parts. First, it has **local power sources**, also called distributed energy resources. These can include solar panels, wind turbines, diesel generators, natural gas generators, fuel cells, or other local generation units. Second, it may include **energy storage**, most commonly batteries, which store extra energy and help support the system when generation changes or demand increases. Third, it has **loads**, which are the devices or buildings that consume electricity, such as lights, computers, motors, labs, houses, or industrial equipment.

Another important part of a microgrid is **power electronics**. Devices such as converters and inverters are needed because some energy sources, like solar panels and batteries, produce or store DC power, while many loads and the main grid use AC power. Power electronics help connect these different parts together and control the flow of energy.

A microgrid also needs a **control system**, which acts like the brain of the system. The controller monitors generation, load demand, battery status, voltage, frequency, and grid conditions. It decides how much power should come from each source, when the battery should charge or discharge, and whether the microgrid should stay connected to the main grid or operate independently.

In a microgrid, different parts may use either AC or DC power, so **converters** are needed to connect them properly. PV panels, batteries, and fuel cells usually produce or store DC, while the main utility grid, most building loads, motors, HVAC systems, and generators usually operate with AC. An inverter converts DC to AC, such as when connecting solar panels or batteries to an AC microgrid. A rectifier converts AC to DC, such as when charging a battery from the grid. A DC/DC converter changes one DC voltage level to another. In simple terms, converters allow AC and DC parts of the microgrid to work together safely at the correct voltage, frequency, and power quality.

The **PCC**, or **Point of Common Coupling**, is the connection point between the microgrid and the main utility grid. It is where the microgrid can connect to the grid, disconnect from it, exchange power, and monitor grid conditions.

2 Modes of operation

One of the most important features of a microgrid is that it can operate in two main modes. In **grid-connected mode**, the microgrid is connected to the main utility grid. In this mode, it can use its own local generation, take extra power from the grid when needed, or sometimes send extra power back to the grid. In **islanded mode**, the microgrid disconnects from the main grid and operates on its own. This can happen during a power outage, fault, emergency, or planned disconnection. In islanded mode, the microgrid must balance its own generation and load because it cannot depend on the main grid.

The **key difference** is that in grid-connected mode, the main grid supports the microgrid by helping regulate voltage and frequency. In islanded mode, the microgrid must manage voltage, frequency, and power balance by itself. This makes control, communication, and battery energy storage more important during islanded operation.

Benefits of micro-grids

Microgrids have several benefits. One major benefit is **reliability and resilience**. If the main grid fails, a microgrid can continue supplying power to important local loads. This is especially useful for hospitals, emergency facilities, campuses, and remote communities. Another benefit is better use of **renewable energy**, because microgrids can combine solar, wind, batteries, and other energy sources in one local system. Microgrids can also improve **energy management** by controlling when to generate, store, or consume energy. In some cases, they can reduce energy costs and make the overall power system more efficient.

Overall, a microgrid is a local, controllable power system that combines generation, storage, loads, power electronics, communication, and control. Its main value is that it can operate with the main grid during normal conditions, but it can also separate and continue operating independently when needed. This makes microgrids important for modern power systems, especially as renewable energy, battery storage, automation, and smart-grid technologies become more common.

AC, DC, and Hybrid Microgrids

Microgrids can be classified based on the type of electrical bus they use: **AC microgrids, DC microgrids, and hybrid AC/DC microgrids**. The bus is the main electrical path where sources, loads, storage, and converters are connected. The choice between AC, DC, or hybrid depends mainly on the type of energy sources, the type of loads, the connection to the main grid, and the overall purpose of the microgrid.

An **AC microgrid** uses an AC bus as its main distribution system. This type is closest to the traditional power grid because the main utility grid is AC, and most buildings are already designed for AC power. AC microgrids are useful when the loads are mainly traditional AC loads, such as motors, HVAC systems, pumps, industrial equipment, lighting systems, and regular building outlets. They are also easier to connect to the main utility grid because both systems operate using AC. However, if the microgrid includes many DC sources such as solar panels or batteries, inverters are needed to convert DC power into AC before connecting them to the AC bus.

A **DC microgrid** uses a DC bus as its main distribution system. This type is useful because many modern energy sources and storage systems are naturally DC. For example, solar PV panels, batteries, and fuel cells produce or store DC power. Also, many modern loads either use DC directly or convert AC to DC internally, such as LED lighting, computers, servers, electronics, EV chargers, and control devices. A DC microgrid can reduce unnecessary power conversions. For example, instead of converting solar PV from DC to AC and then back to DC for a DC load, the power can stay on the DC side. This can improve efficiency and simplify the connection of PV, batteries, and electronic loads. However, DC microgrids are less directly compatible with the AC utility grid and traditional AC loads, so converters are still needed when connecting to AC systems.

A **hybrid AC/DC microgrid** contains both an AC bus and a DC bus. These two buses are connected using a bidirectional converter, which allows power to flow between the AC and DC sides. This structure is very flexible because AC sources and loads can stay on the AC side, while DC sources and loads can stay on the DC side. For example, the AC side may include the utility grid, diesel generators, motors, HVAC systems, and building loads, while the DC side may include solar panels, batteries, EV chargers, and electronic loads. Hybrid microgrids are often practical because real systems usually contain both AC and DC equipment. The main advantage is flexibility, but the main challenge is that the control system, protection system, and power management become more complex.

In simple terms, an **AC microgrid** is best when the system mainly serves traditional AC loads and needs easy connection to the main grid. A **DC microgrid** is best when the system has many DC sources and DC loads, such as solar PV, batteries, EV chargers, electronics, or data centers. A **hybrid microgrid** is best when the system has a mix of AC and DC components and needs the advantages of both. For modern microgrid design, hybrid systems are often attractive because they can combine renewable energy, battery storage, traditional AC loads, and modern DC loads in one flexible system.

Simple comparison

Type	Main bus	Best for	Main advantage	Main challenge
AC microgrid	AC bus	Buildings, motors, grid connection	Easy grid connection, familiar system	More conversions for PV/battery
DC microgrid	DC bus	PV, batteries, EVs, electronics, data centers	Fewer conversions for DC systems	Harder connection to AC grid/AC loads
Hybrid microgrid	AC + DC buses	Mixed AC and DC systems	Most flexible and practical	More complex control and protection

Data Monitoring in a Microgrid

In a microgrid, monitoring is important because the operator needs to understand what is happening in the system in real time. The dashboard should not only show whether the microgrid is producing power, but also where the power is coming from, where it is going, whether the battery is charging or discharging, whether the voltage and frequency are stable, and whether the microgrid is connected to or disconnected from the main grid. The exact data depends on whether the microgrid is AC, DC, or hybrid, but the main idea is the same: monitor generation, storage, loads, grid connection, converter status, and communication health.

What kind of data should be monitored?

The main data to monitor includes **PV power, wind power, load demand, battery state of charge, battery charging/discharging power, bus voltage, current, frequency, total generated power, power imported or exported from the grid, breaker status, converter/inverter status, communication delay, packet loss, and alarms or abnormal behavior**.

For an AC microgrid, important values include AC voltage, current, frequency, active power, reactive power, and power factor. Frequency is important because AC systems must stay close to their rated frequency, such as 60 Hz in North America. For a DC microgrid, the main electrical value is usually DC bus voltage, because there is no frequency in a DC system. For a hybrid AC/DC microgrid, both AC-side and DC-side data should be monitored.

Where is the data monitored?

Data is monitored at important points in the microgrid. At the **PV system**, the output power, voltage, and current can be measured to know how much solar energy is being produced. At the **wind turbine**, power, voltage, current, and sometimes wind speed or turbine speed can be monitored. At the **battery**, the battery management system can provide state of charge, voltage, current, temperature, and charging or discharging power.

Data should also be monitored at the **loads**, because the operator needs to know how much power is being consumed. For example, building loads, motors, HVAC systems, labs, or industrial equipment may be monitored individually or as a group. Important buses should also be monitored, such as the **AC bus**, **DC bus**, or both in a hybrid system. These bus measurements help show whether voltage, current, and power flow are stable.

Another very important location is the **PCC** or Point of Common Coupling. This is where the microgrid connects to the main utility grid. At the PCC, the system can monitor grid voltage, current, frequency, imported power, exported power, breaker status, and whether the microgrid is operating in grid-connected mode or islanded mode.

How is the data monitored?

The data is monitored using sensors, meters, controllers, and communication systems. Voltage sensors or potential transformers measure voltage. Current sensors or current transformers measure current. From voltage and current, the system can calculate active power, reactive power, apparent power, power factor, and energy consumption. Inverters, converters, relays, and battery management systems can also send internal data such as operating status, faults, temperature, and control mode.

This measured data is sent to a controller or monitoring system using communication protocols. In real systems, this may involve industrial protocols such as **Modbus**, **DNP3**, **IEC 61850**, or other communication methods. The controller or data acquisition system collects the information and sends it to a **dashboard/GUI**, where the operator can see the condition of the microgrid in a clear way.

Why is this data monitored?

The data is monitored to make sure the microgrid is operating safely, reliably, and efficiently. Monitoring PV and wind power helps the operator know how much renewable energy is available. Monitoring load demand shows how much power is being consumed. Monitoring the battery state of charge helps decide when the battery should charge or discharge. Monitoring bus voltage and frequency helps detect instability or poor power quality. Monitoring the PCC shows whether the microgrid is importing power from the grid, exporting power to the grid, or operating independently in islanded mode.

Communication data is also important. If there is a communication delay, packet loss, or missing data, the dashboard may not show the real condition of the microgrid correctly. This can affect control decisions and operator awareness. Later, AI could be added to detect abnormal behavior, equipment faults, cyberattack-related behavior, or communication problems.

Summary

In a microgrid, the dashboard should monitor data from the generation sources, battery, loads, buses, converters, and PCC. Important values include PV power, wind power, load demand, battery state of charge, voltage, current, frequency, grid import/export power, breaker status, converter status, and communication performance. This data is collected using sensors, meters, inverters, controllers, and communication protocols, then displayed to the operator through a dashboard. The purpose of monitoring is to help the operator understand the real-time condition of the microgrid, maintain stable operation, detect problems, and make better control decisions.

Role of MATLAB/Simulink in Microgrid Modeling

MATLAB/Simulink is an important tool for microgrid research because it allows the physical electrical system to be modeled and tested in simulation before building or testing it in real life. Instead of working directly with hardware, a virtual microgrid can be created using block diagrams. Each block represents a real component in the system, such as a PV panel, wind turbine, battery, inverter, converter, load, controller, breaker, bus, or grid connection. This makes it possible to study how the microgrid behaves under different operating conditions in a safe and flexible way.

For microgrid modeling, Simulink is often used with **Simscape Electrical**, which provides physical electrical components that can be connected similarly to a real circuit or power system. For example, a PV array block can represent solar generation, a battery block can represent energy storage, an inverter can convert DC power to AC power, and loads can represent the power demand from buildings, motors, HVAC systems, or other equipment. The user can enter important parameters such as PV power rating, battery voltage and capacity, load size, converter ratings, grid voltage, frequency, and control settings. After these values are entered, the system can be simulated to observe its behavior.

Simulink works like a virtual laboratory for testing microgrid operation. Different scenarios can be studied, such as solar power increasing or decreasing, wind power changing, load demand rising, the battery charging or discharging, or the microgrid switching between grid-connected and islanded mode. This is useful because the researcher can see how the system responds before applying the design in a real system. If the voltage becomes unstable, the battery discharges too quickly, or the load is not supplied properly, the design or control strategy can be adjusted and tested again.

Measurement blocks can also be added throughout the Simulink model. These blocks can measure important data such as bus voltage, current, active power, reactive power, frequency, battery state of charge, battery charging/discharging power, load demand, and power imported from or exported to the grid. This measured data can be viewed inside Simulink using scopes, saved to MATLAB for analysis, or exported to another tool. This is important because the same type of data would later be shown on a dashboard for an operator.

For this research project, Simulink can be used mainly to represent the **physical electrical part** of the microgrid. A communication simulator can then represent how the measured data is transferred between devices, controllers, and the dashboard. In this setup, Simulink models the actual electrical behavior of the microgrid, while the communication simulator studies the data transfer side, such as communication delay, packet loss, or possible abnormal communication behavior. The dashboard can then display the received data clearly to the operator.

In simple terms, Simulink is where the microgrid is built virtually, tested under different conditions, measured, and analyzed. It helps connect the electrical design side of the project with the communication and dashboard side, making it a useful tool for studying modern microgrid systems.

Communication system basics in a microgrid

In a microgrid, the electrical system produces physical data, but the operator does not directly “see” electricity flowing. The data must be measured, transmitted, processed, and displayed. So, the communication system connects the physical electrical layer to the control and monitoring layer.

Simple flow:

Sensors/meters → communication network → controller/SCADA → dashboard/operator

For example, a voltage sensor measures bus voltage, a smart meter measures power, the battery management system sends battery state of charge, and an inverter sends its operating status. This data is transmitted through a communication network to a controller or SCADA system. The controller may then send commands back, such as changing inverter output, charging/discharging the battery, opening/closing a breaker, or switching operating mode.

NIST describes operational technology systems as systems that monitor or control physical devices and processes, which matches the role of communication and control in microgrids.

Main devices/terms to know

Sensor: Measures a physical/electrical quantity, such as voltage, current, temperature, frequency, or power.

Smart meter: Measures electrical energy or power data and can communicate that information digitally.

PLC: This can mean two things, so be careful. In control systems, PLC usually means Programmable Logic Controller, a controller used to automate equipment. In communication, PLC can also mean **Power Line Communication**, where data is sent through power lines.

RTU: A **Remote Terminal Unit** collects data from field devices and sends it to a central control system. It can also receive commands from the control system.

IED: An **Intelligent Electronic Device** is a smart power-system device, often used for protection, control, metering, or monitoring. Examples include digital relays, smart breakers, and advanced meters.

SCADA: Supervisory Control and Data Acquisition. This is the system used to collect data from the field, monitor operation, display system status, and sometimes send supervisory commands. NIST lists SCADA, PLCs, and related control systems as major parts of industrial control systems.

Communication protocol: The rule/language that devices use to exchange data. Examples in power and industrial systems include Modbus, DNP3, IEC 61850, TCP/IP, and others.

Latency/delay: The time it takes for data to travel from one point to another. For example, the delay between a sensor measuring voltage and the controller receiving that voltage.

Packet loss: When some transmitted data does not arrive correctly. This matters because missing data can affect monitoring and control decisions.

Data rate: How much data can be transferred per second.

Cyber vulnerability: A weakness in the communication or control system that could be attacked, causing false data, missing data, wrong commands, or system disruption.

Why communication matters

The communication system is important because modern microgrids are not only electrical systems; they are also cyber-physical systems. The physical side includes PV panels, batteries, converters, loads, buses, and PCC. The cyber/communication side transfers measurements and commands between devices, controllers, and operators.

For your project, the idea is probably:

Simulink models the physical electrical microgrid.
A communication simulator models how the measurement data is transmitted.
A dashboard displays the received data to the operator.

So, the electrical model may produce values like PV power, battery SOC, voltage, current, frequency, and load demand. The communication model can then study whether this data reaches the dashboard correctly, whether there is delay, whether packets are lost, and whether abnormal communication behavior occurs.

Summary

The **electrical system** creates real operating data.
The **communication system** transfers that data.
The **controller/SCADA** processes and manages that data.
The **dashboard** displays it to the operator.
The **communication simulator** is used to test this data-transfer process virtually.

Communication Simulator

A **communication simulator** is software used to model how data moves inside a microgrid communication network.

In our project, Simulink may simulate the **physical electrical system**, like PV power, battery SOC, bus voltage, and load demand. Then a communication simulator can simulate how that measured data is sent from sensors/meters/controllers to the dashboard or operator.

It can help test things like:

- how long data takes to arrive, called **delay/latency**
- whether some data is lost, called **packet loss**
- how much data the network can handle
- how different devices communicate
- what happens during faults, congestion, or cyberattack-like behavior

Simple idea:

Simulink = simulates the electrical microgrid
Communication simulator = simulates the data-transfer network
Dashboard = displays the received data

So, the communication simulator does not generate the electrical power itself. It only models the communication side: how measurements and commands travel between devices.

DC Microgrid Test Simulations for learning Under Different Conditions

Constant Load and Source (simple DC circuit) Simulation

For the first simulation, I built a simple DC grid model in MATLAB/Simulink using Simscape physical electrical components. The model consisted of a 48 V DC voltage source connected to a DC bus and a constant 10 Ω resistive load. Voltage and current sensors were added to measure the load voltage and current, and a Product block was used to calculate the load power. This simulation was done to verify that the basic DC circuit, measurement blocks, and Simscape connections were working correctly before adding more complex microgrid components. The results matched the expected values: the voltage stayed at 48 V, the current was 4.8 A, and the power was 230.4 W. This confirmed that the basic DC source–bus–load model was working properly.

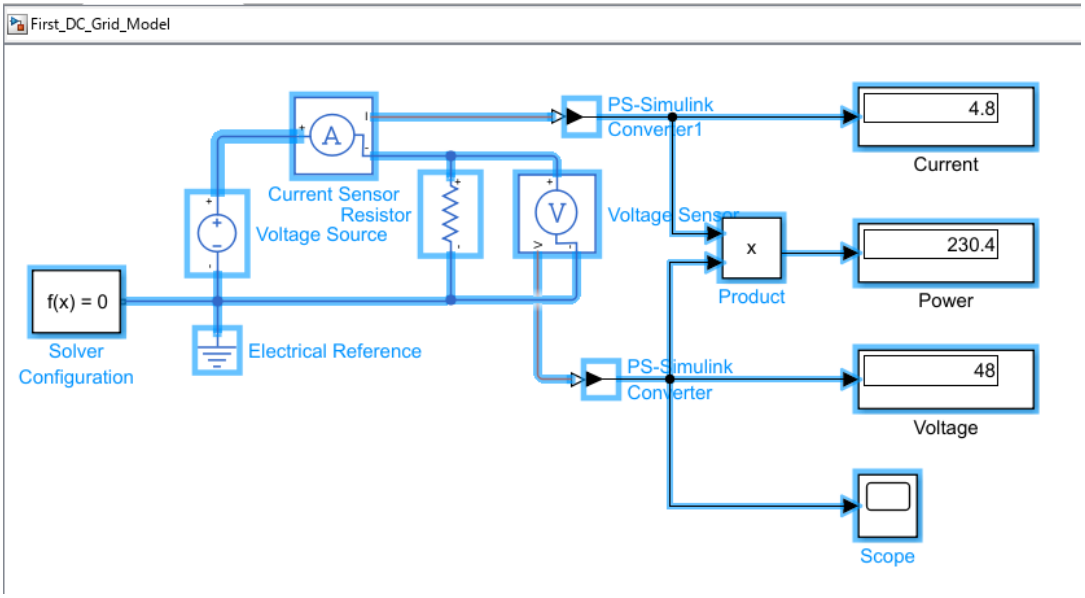


Fig (1): Simulink Schematic (Constant Load)

Variable Load Simulation

For the second simulation, the fixed resistor was replaced with a variable resistor to represent a changing load demand on the DC bus. A Step block was used to change the resistance from 10 Ω to 5 Ω at 0.5 seconds, and the signal was connected to the variable resistor through a Simulink-PS Converter. When the resistance decreased, the load current increased from 4.8 A to 9.6 A, and the power increased from 230.4 W to 460.8 W. The bus voltage remained at 48 V because the source was modeled as an ideal DC voltage source. This simulation showed how a DC grid responds when the load demand suddenly increases, and it helped me understand the relationship between load resistance, current, and power before moving on to more realistic sources such as PV panels, batteries, or wind systems.

NOTE: In the following simulink figure the displays only show the final values.

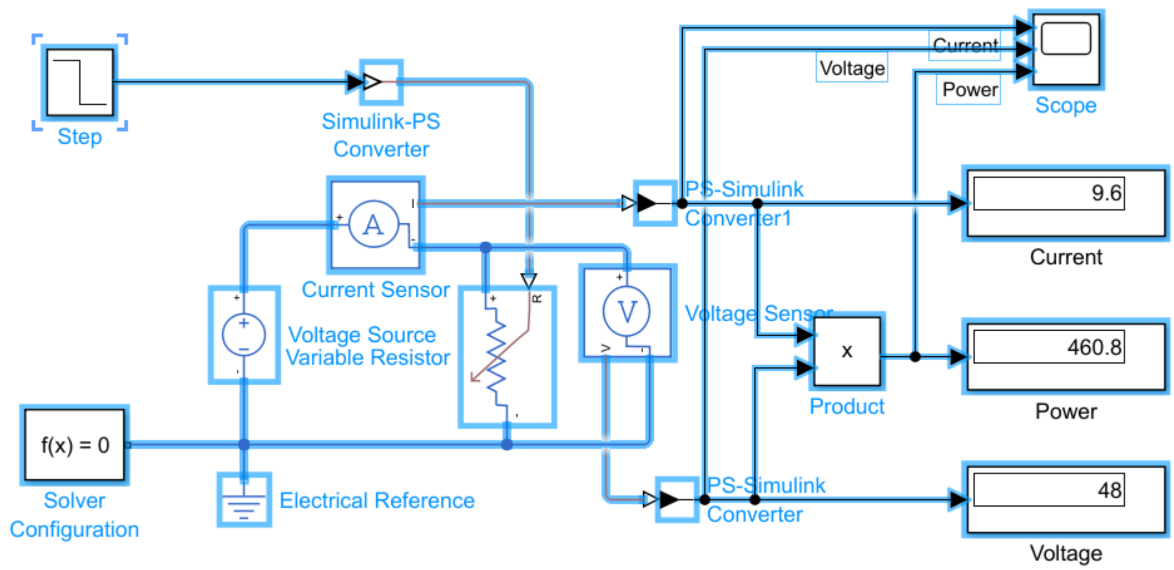


Fig (2): Simulink Schematic for Variable Load

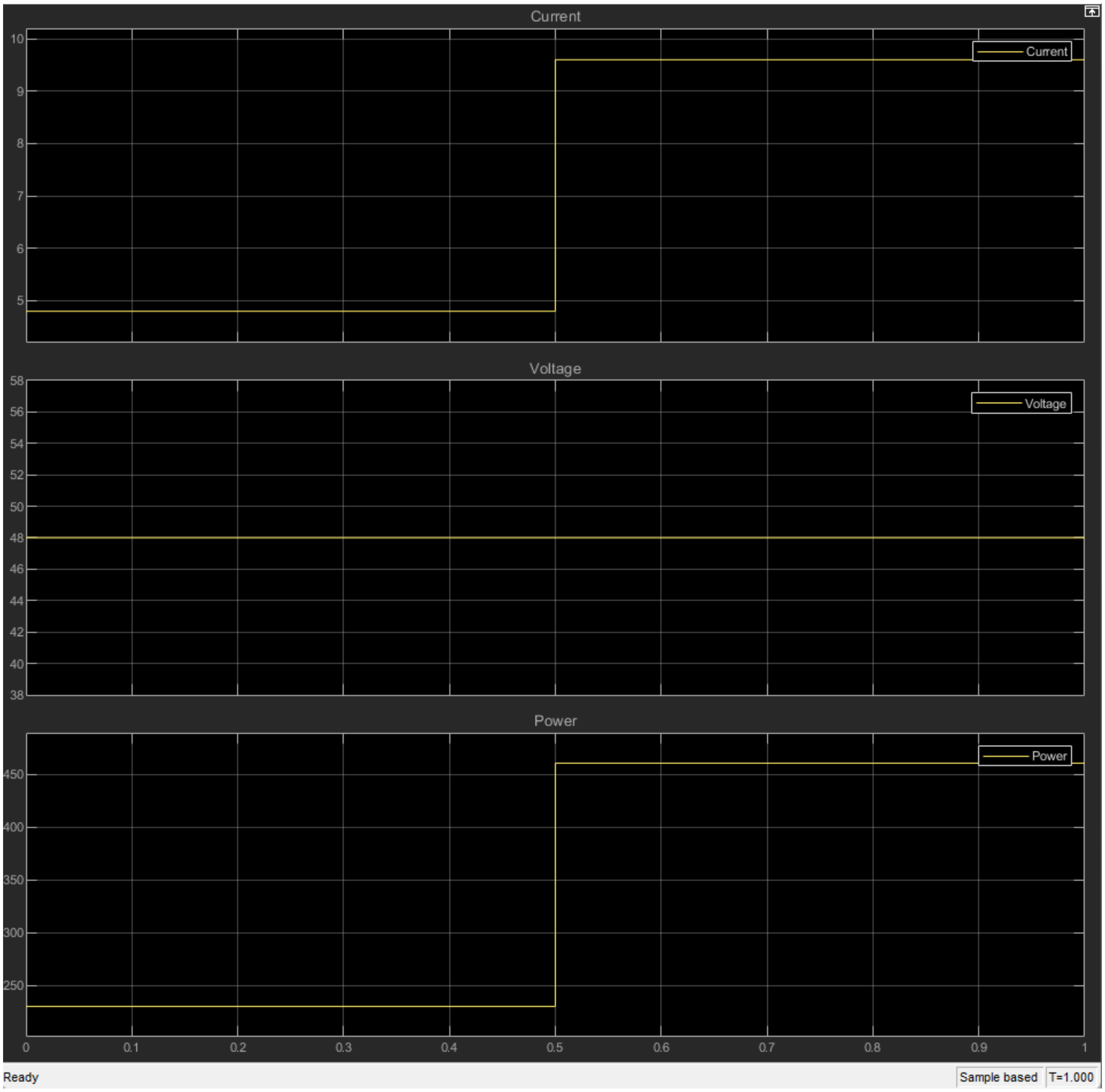


Fig (3): Variable Load Simulation Results

Variable DC Voltage Source with Constant Load Simulation

For this simulation, I tested the effect of changing the source voltage while keeping the load constant. The main objective was to study how the DC bus, load current, and load power respond when the supply side changes. In the model, the fixed DC voltage source was replaced with a controlled voltage source, and a Step block was used to change the source voltage from 48 V to 36 V at 0.5 seconds. The load was kept constant at 10 Ω, so any change in current and power was caused only by the change in the source voltage.

The goal of this simulation was to represent a simple version of a variable energy source, such as a PV source affected by changing sunlight or a wind system affected by changing wind speed. The results matched the expected behavior. Before 0.5 seconds, the voltage was 48 V, the current was 4.8 A, and the power was 230.4 W. After the voltage dropped to 36 V, the current decreased to 3.6 A and the power decreased to 129.6 W. This confirmed that when the source voltage decreases, the load current and delivered power also decrease. This simulation helped me understand the source-side behavior of a DC grid before replacing the simple controlled source with more realistic sources such as PV panels, batteries, or wind generation systems.

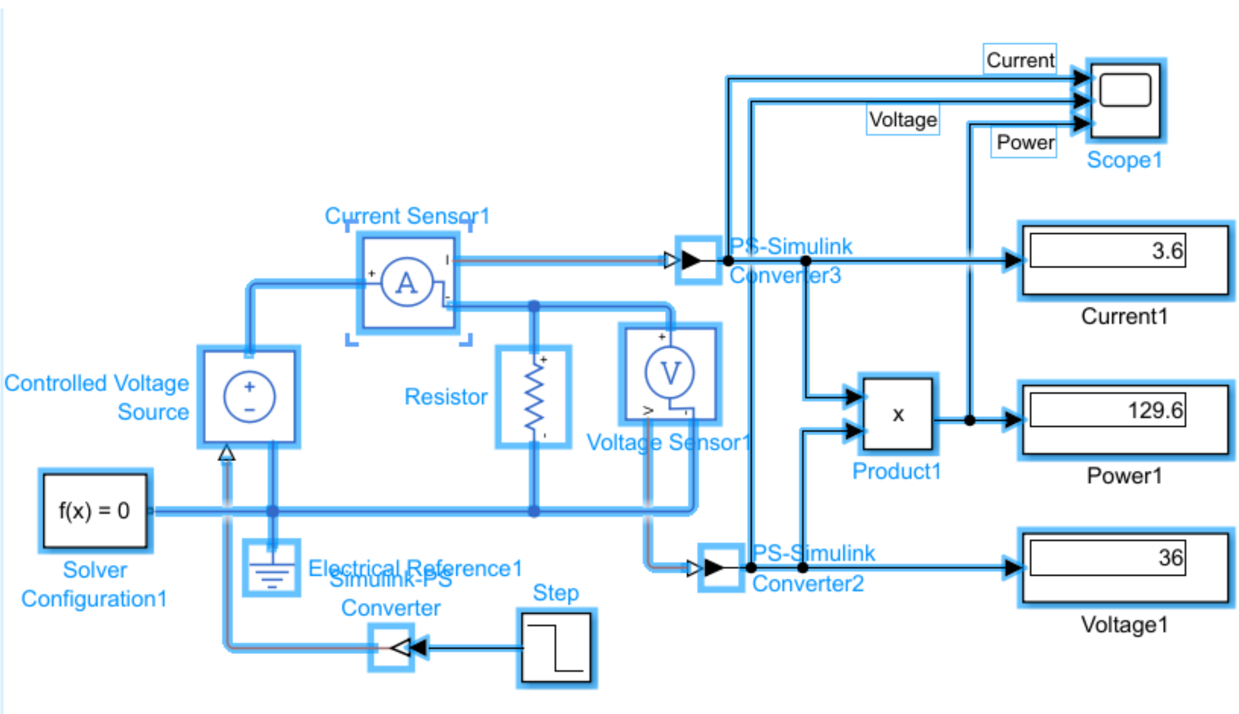


Fig (4): Simulink Schematic for Variable Voltage Source – Constant Load Simulation

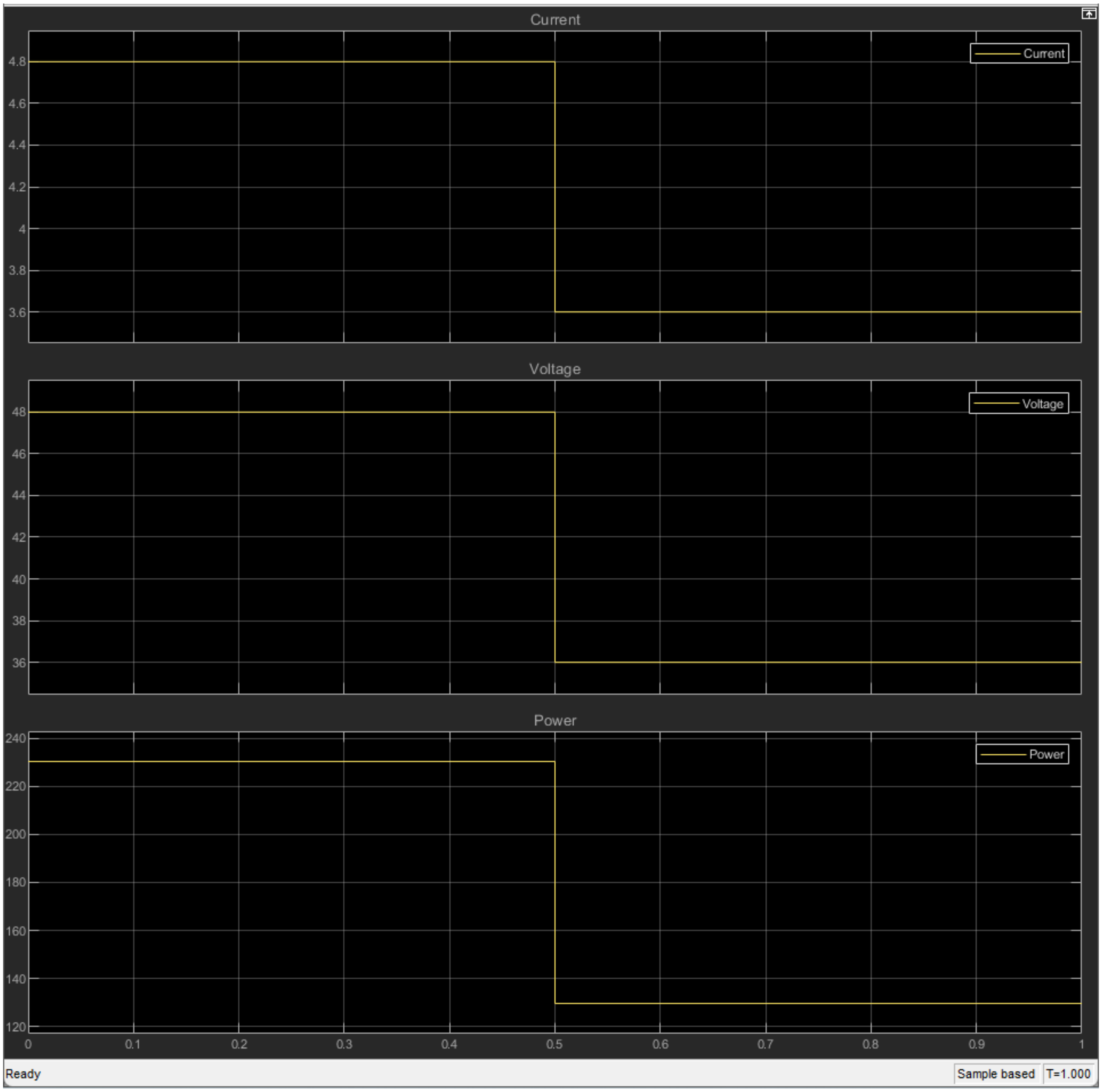


Fig (5): Variable Voltage Source – Constant Load Simulation Results

Variable DC Current Source with Constant Load Simulation

In this simulation, I replaced the ideal DC voltage source with a controlled current source to create a simple PV-like source model. This was done because a photovoltaic source is closer to a current-producing source than a perfect voltage source, since its output current changes with sunlight level. A Step block was used to control the current source, changing the injected current from 5 A to 3 A at 0.5 s. The load resistance was kept constant at 10 Ω. Before the step change, the current was 5 A, giving a load voltage of 50 V and a power of 250 W. After the step change, the current decreased to 3 A, so the load voltage dropped to 30 V and the power dropped to 90 W. This simulation showed how a change in PV-like source current affects the DC bus voltage and load power. It also helped connect the basic Simscape measurement setup to a more realistic microgrid idea, where source power can vary depending on operating conditions such as sunlight.

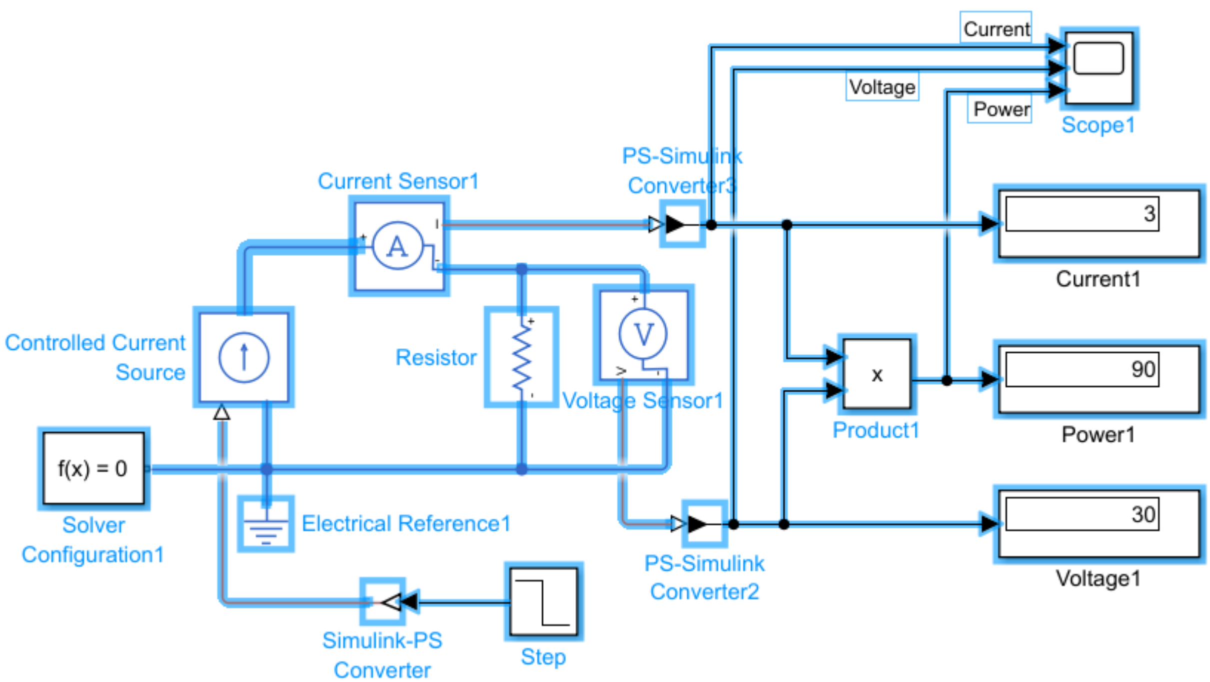


Fig (6): Simulink Schematic for Variable Current Source – Constant Load Simulation

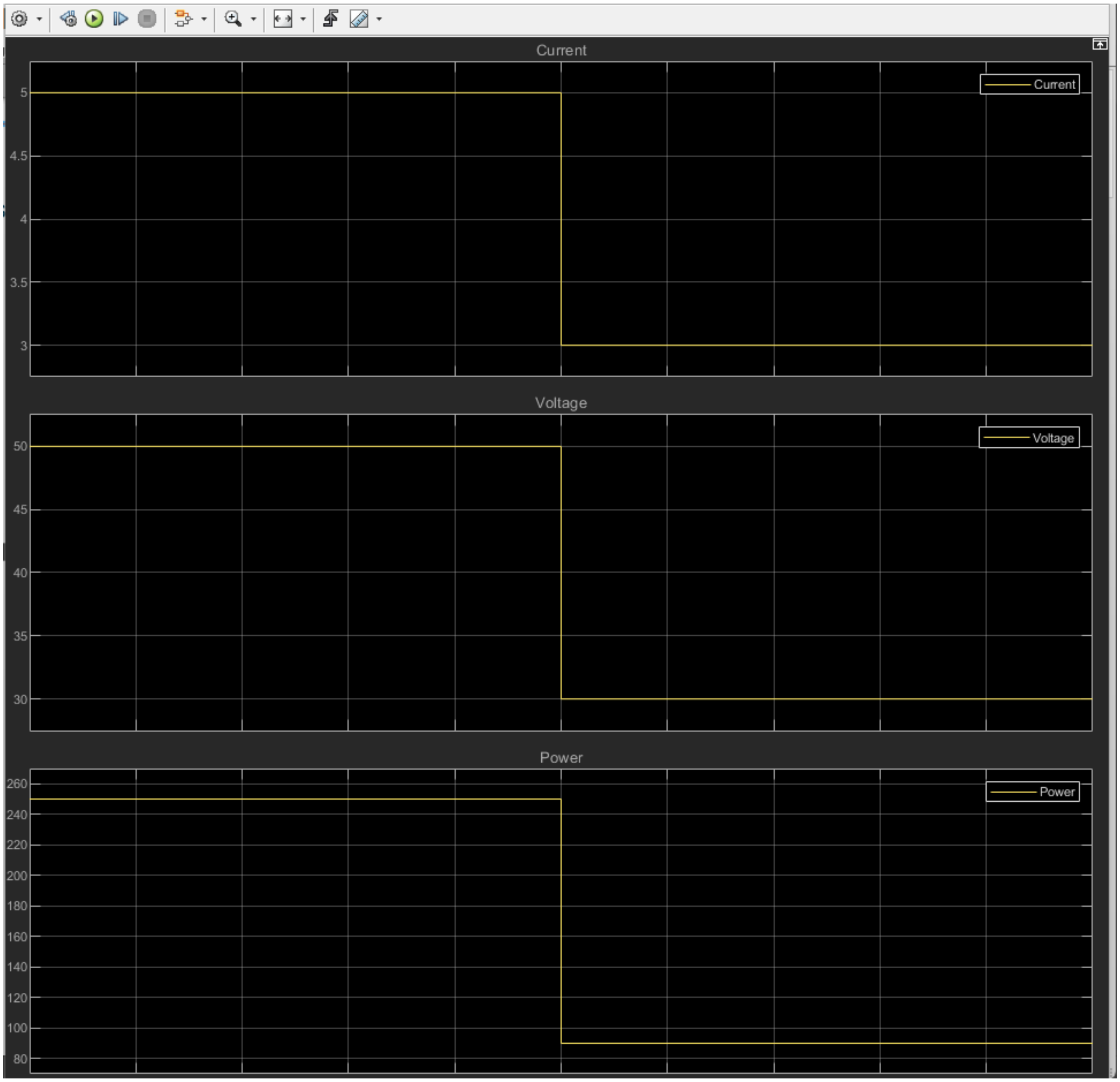


Fig (7): Variable Current Source – Constant Load Simulation Results

Variable DC Current Source with Constant Load and a capacitor Simulation

In this simulation, a DC bus capacitor was added in parallel with the load to study the dynamic behavior of the DC bus voltage. In a real DC microgrid, capacitors are commonly used on the DC bus to store energy and reduce sudden voltage changes caused by source or load variations. The controlled current source was used as a simplified PV-like source, and its current was changed from 5 A to 3 A at 0.5 s while the load resistance was kept at 10 Ω. At the beginning, the voltage increased gradually because the capacitor was charging toward the steady-state bus voltage of 50 V. When the current decreased, the bus voltage did not drop instantly; instead, it slowly decreased toward the new steady-state value of 30 V as the capacitor discharged. The power followed the same trend because it depends on both voltage and current. This simulation showed that adding a DC bus capacitor smooths the voltage response and makes the model closer to a real DC microgrid system.

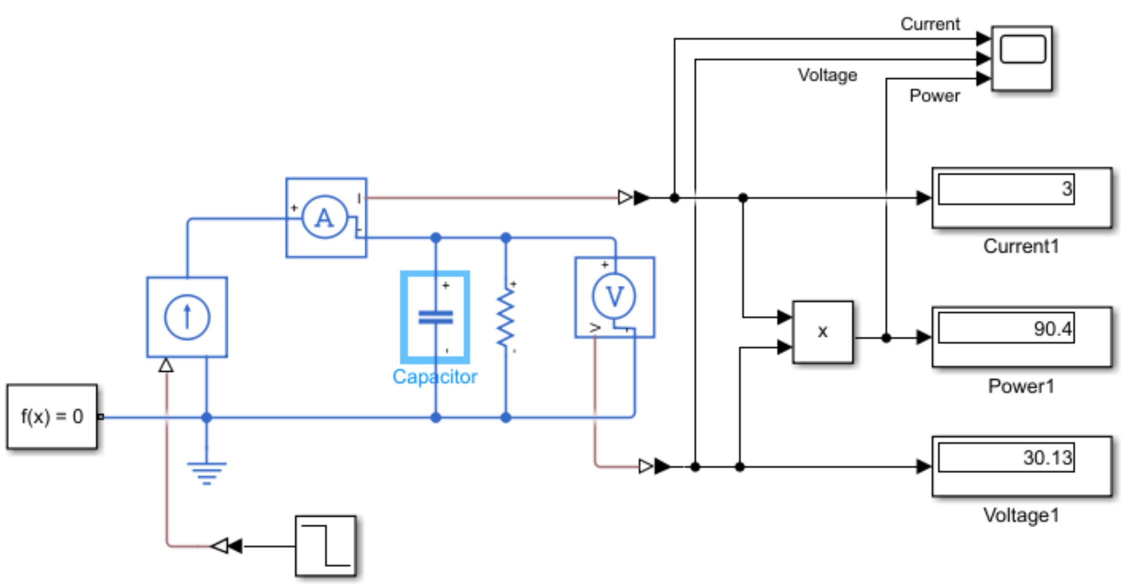


Fig (8): Simulink Schematic for Variable Current Source – Constant Load and capacitor Simulation

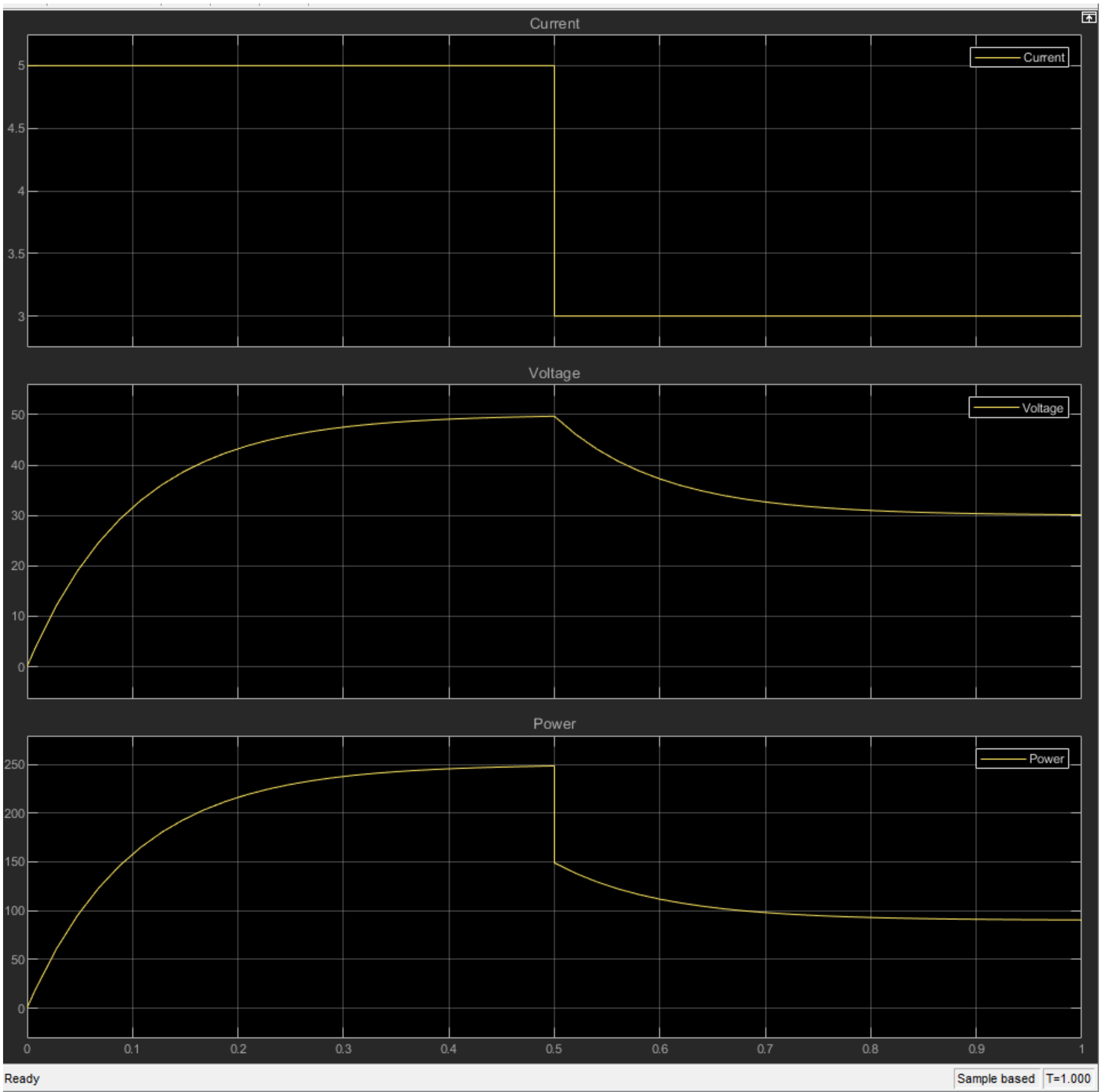


Fig (9): Variable Current Source – Constant Load and capacitor Simulation Results

Use of Subsystems

I implemented the PV-like source as a **subsystem** to make the main Simulink model cleaner, more organized, and easier to understand. By grouping the source components into one block, the overall circuit layout becomes less cluttered and easier to modify or debug. This structure also makes the model more flexible, because additional subsystems or different types of energy sources can be added in the future without changing the main model significantly.

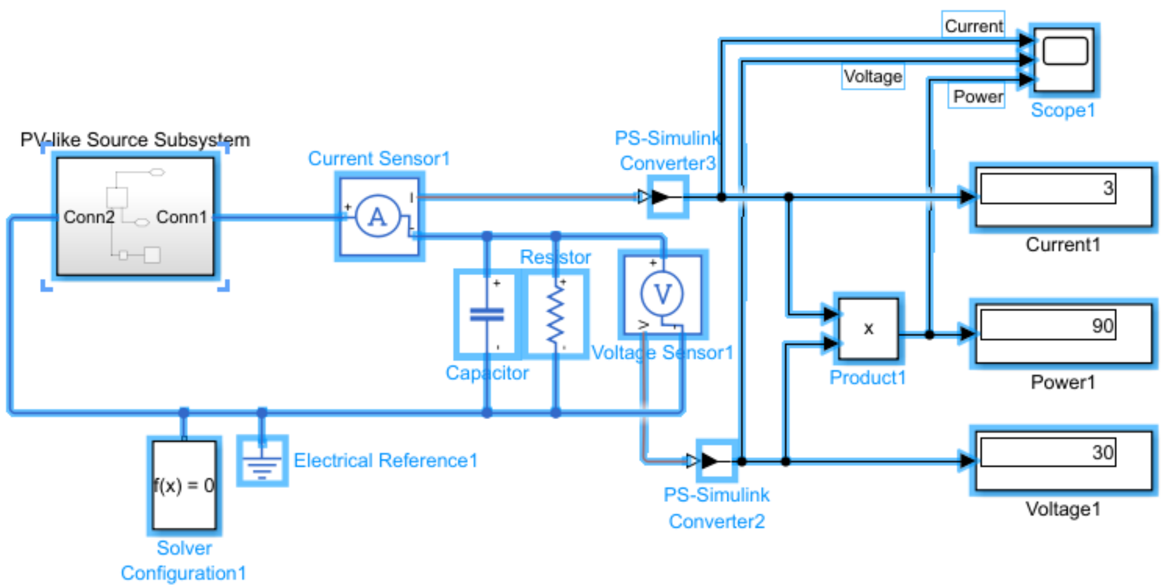


Fig (10): Subsystem Schematic

DC-DC Boost Converter Test

In this simulation, a DC-DC converter was added between the input source and the DC bus to test voltage conversion and power transfer. The purpose of this test was to move beyond a direct source-to-load connection and begin modeling a more realistic power-electronics-based DC grid. A 30 V DC source was connected to the input side of the converter, and the converter output was regulated to 50 V. The output side included a 10 Ω resistive load, a DC bus capacitor, voltage and current sensors, and power measurement blocks.

The results showed that the converter successfully regulated the output bus voltage to 50 V. With a 10 Ω load, the output current was 5 A and the output power was 250 W. The input side measured about 30 V, 8.7 A, and 261 W. The input power was slightly higher than the output power because the converter was modeled with efficiency losses. This confirmed that the DC-DC converter can step up and regulate the DC bus voltage while still following power balance, meaning it changes the voltage/current ratio but does not create energy. This test is important because future PV, battery, and wind branches will need converters before connecting to the common DC bus.

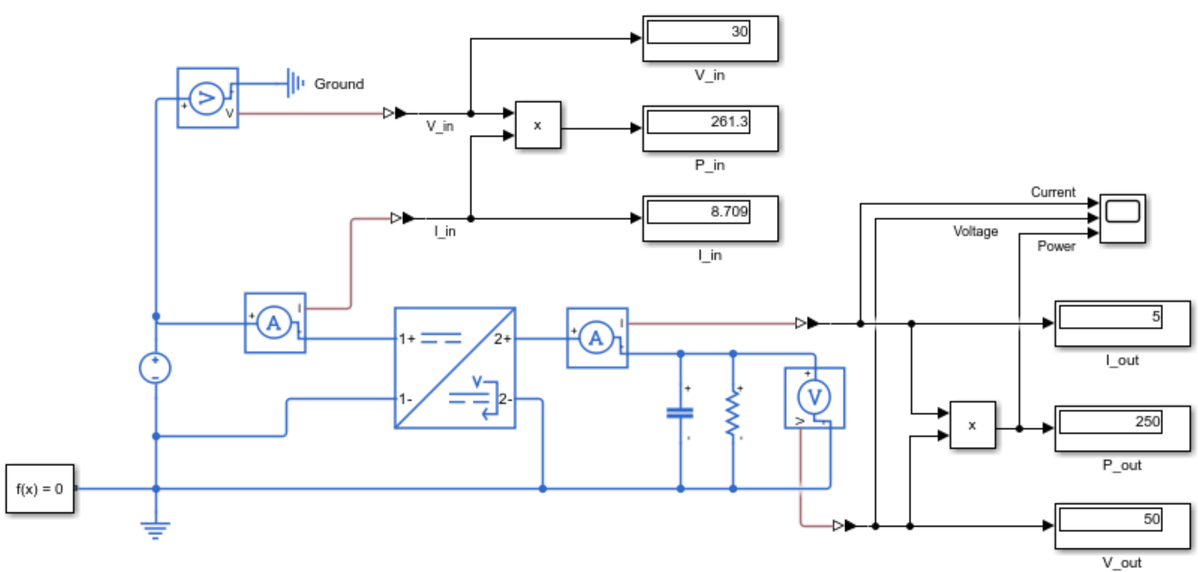


Fig (11): DC-DC converter test Schematic

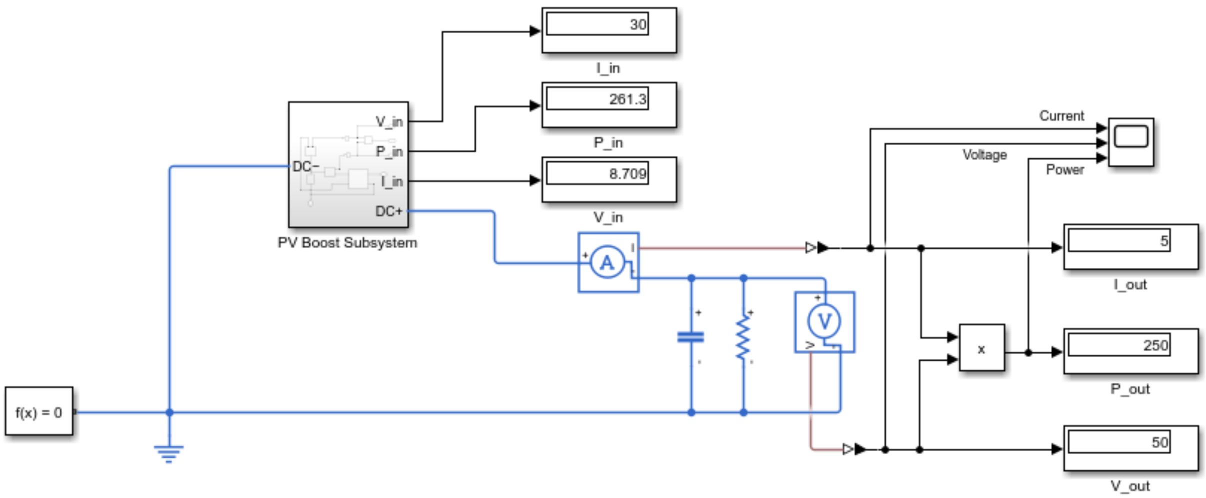


Fig (12): Conversion to Subsystem schematic

Actual PV Subsystem Test with Buck-Boost Converter

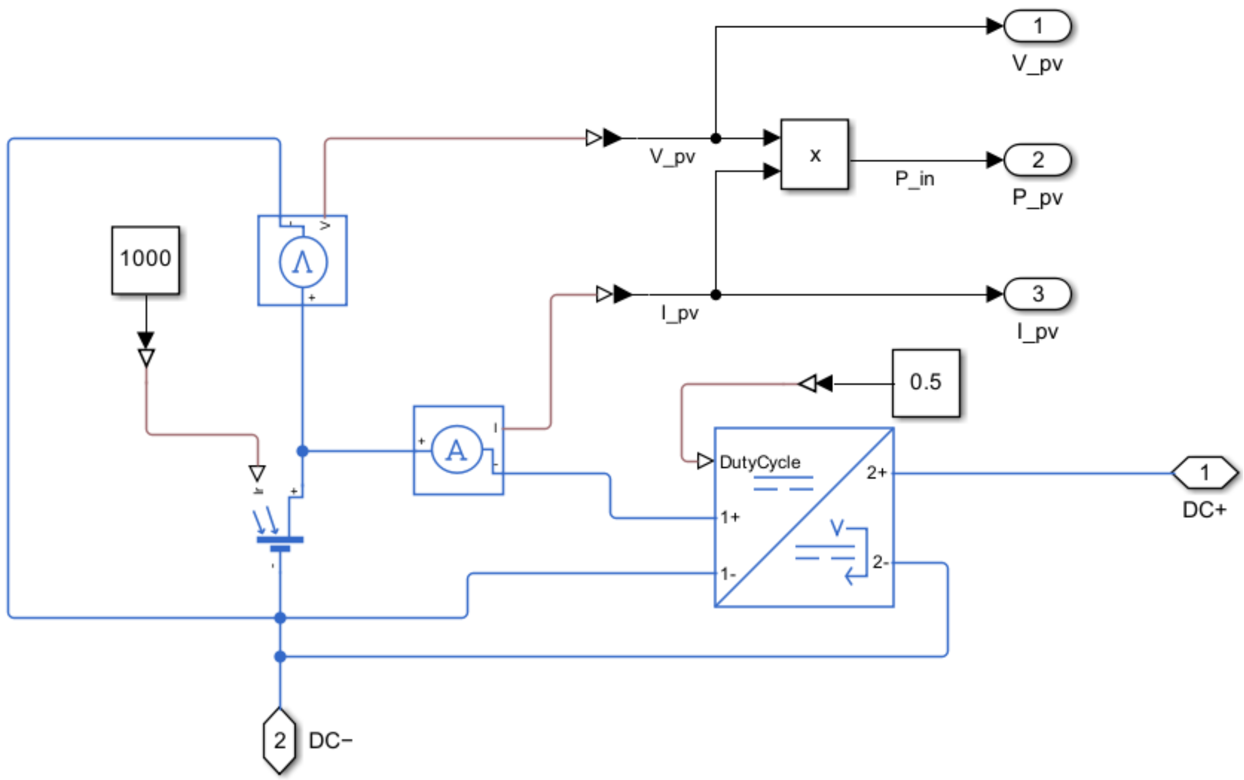
In this test, the fixed DC voltage source was replaced with an actual Solar Cell/PV model. An irradiance input was added using a Constant block set to 1000 W/m^2 and connected through a Simulink-PS Converter to the PV irradiance port. This made the source behave more like a real PV panel affected by sunlight.

The previous boost converter was replaced with an Average-Value DC-DC Converter configured as a buck-boost converter. A fixed duty-cycle input was added temporarily to control the converter. Later, this fixed duty cycle can be replaced by an MPPT controller.

PV voltage, current, and power were measured using sensors and displayed outside the subsystem. After adjusting the PV model to use multiple series cells, the output became realistic. A small startup transient appeared because the bus capacitor was charging and the converter was settling, then the system reached steady-state operation. This confirmed that the actual PV source, irradiance input, buck-boost converter, and measurements were working correctly.

Building Main DC Micro-Grid Model

PV subsystem schematic



In this test only the subsystem was changed, the rest of the circuit is the same.

Local MPPT Controller for PV Subsystem

Objective

The goal of this step was to make the PV subsystem more realistic by replacing the fixed duty-cycle input with a local MPPT controller. Instead of manually setting the duty cycle, the controller automatically adjusts it to extract the maximum available power from the PV panel.

Components Added

The following control components were added to the PV subsystem:

- MATLAB Function block for the MPPT algorithm
- Zero-Order Hold blocks for sampled PV voltage and current signals
- Saturation block to limit the duty cycle between 0.05 and 0.95
- Unit Delay block to avoid an algebraic loop
- Simulink-PS Converter to send the duty-cycle signal to the buck-boost converter
- Display and Scope to monitor the duty cycle over time

Controller Operation

The MPPT controller uses the measured PV voltage and current to calculate PV power:

$P_{pv} = V_{pv} \times I_{pv}$

It then uses a Perturb and Observe method. The controller slightly changes the duty cycle and checks whether the PV power increases or decreases. If the power increases, it keeps changing the duty cycle in the same direction. If the power decreases, it reverses direction.

Results

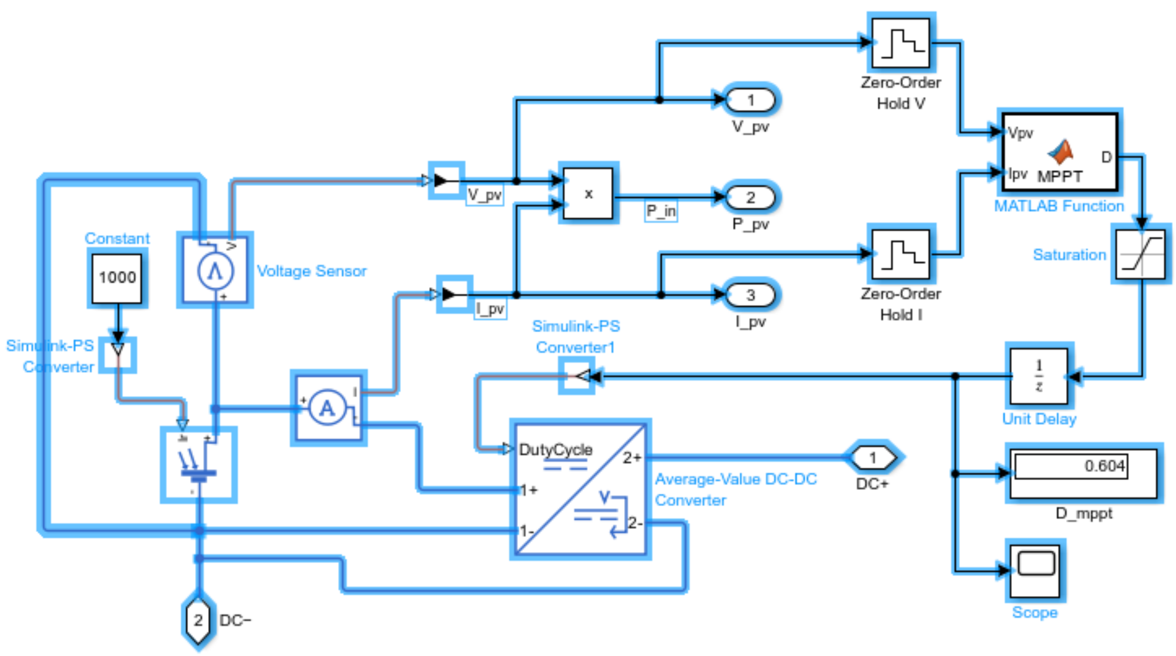
After adding the MPPT controller, the duty cycle started near 0.5 and gradually settled around 0.604. The small oscillation around this value is normal because the Perturb and Observe method continuously tests small duty-cycle changes around the maximum-power point.

The PV power increased from about 119 W with the fixed duty cycle to about 203.6 W with MPPT. This confirmed that the controller successfully improved the PV operating point and extracted more power from the PV source.

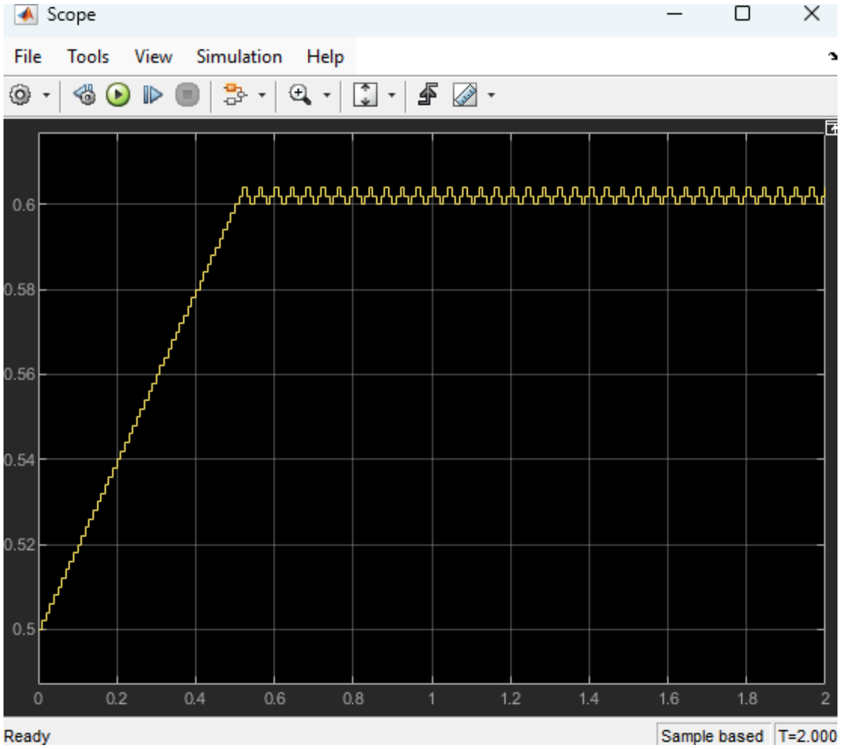
Conclusion

This step completed the local control stage of the PV subsystem. The PV subsystem now includes an actual PV model, irradiance input, buck-boost converter, measurement outputs, and a local MPPT controller that automatically adjusts the duty cycle to maximize PV power.

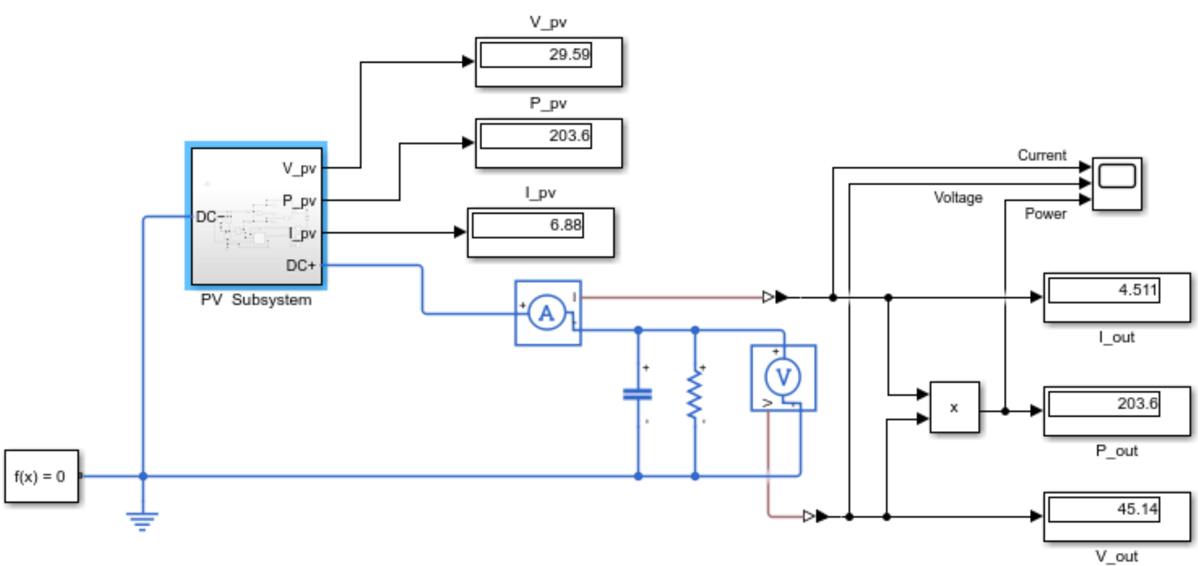
Schematic for Local MPPT Controller for PV Subsystem



Scope for Duty Cycle



Whole Model

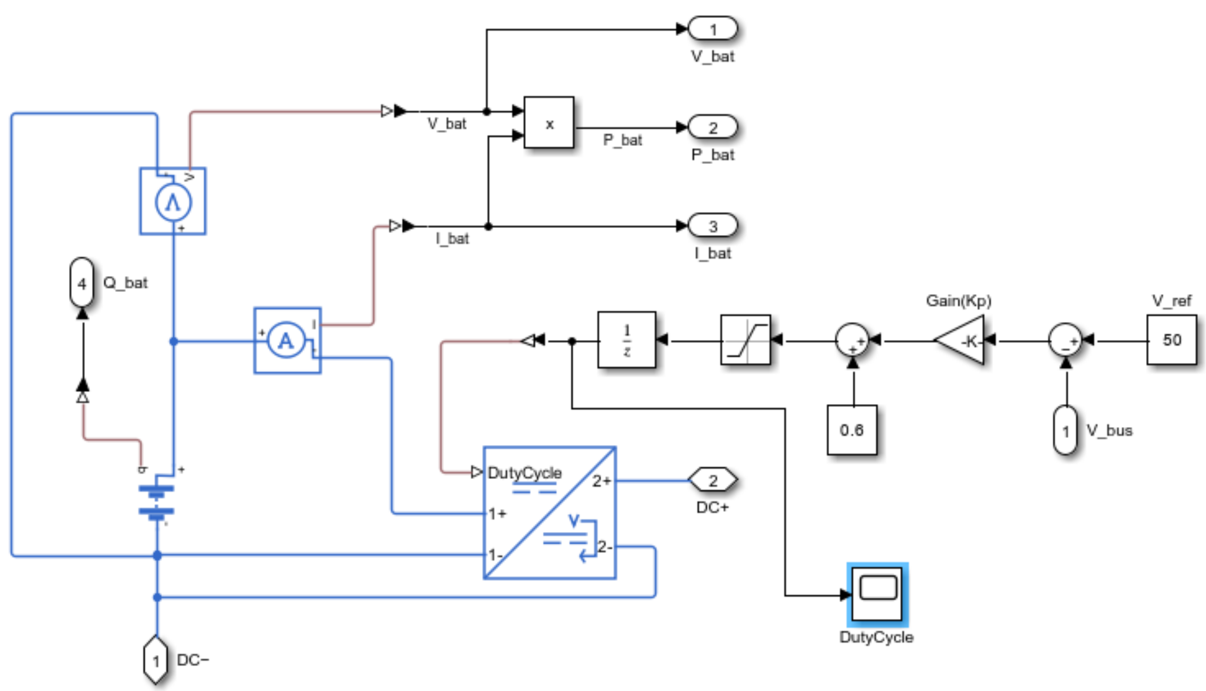


Battery Subsystem

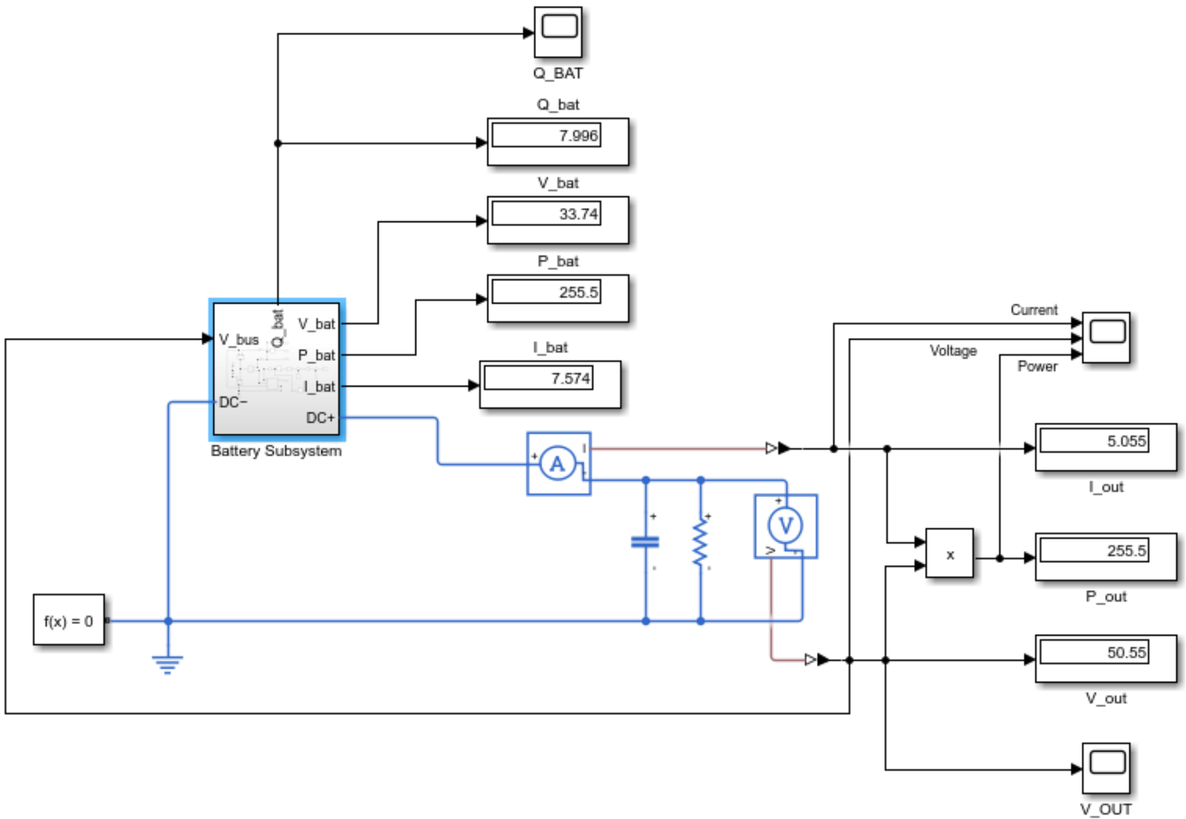
Battery Subsystem Components

The battery subsystem was built as a separate source branch for the DC microgrid. Inside the subsystem, I added a Battery block with a nominal voltage of 36 V and a capacity of 10 Ah. The initial charge was set to 8 Ah, which represents about 80% charge. Voltage and current sensors were added to measure the battery-side voltage and current, and the battery power was calculated using the measured voltage and current. I also enabled the battery charge measurement port and exposed it as Q_bat so the battery charge could be monitored during the simulation. An Average-Value DC-DC Converter configured as a buck-boost converter was added between the battery and the DC bus so the battery voltage could be stepped up to support the 50 V DC bus.

Fig (13): Battery subsystem with local controller schematic



Whole Circuit



Local Battery Controller

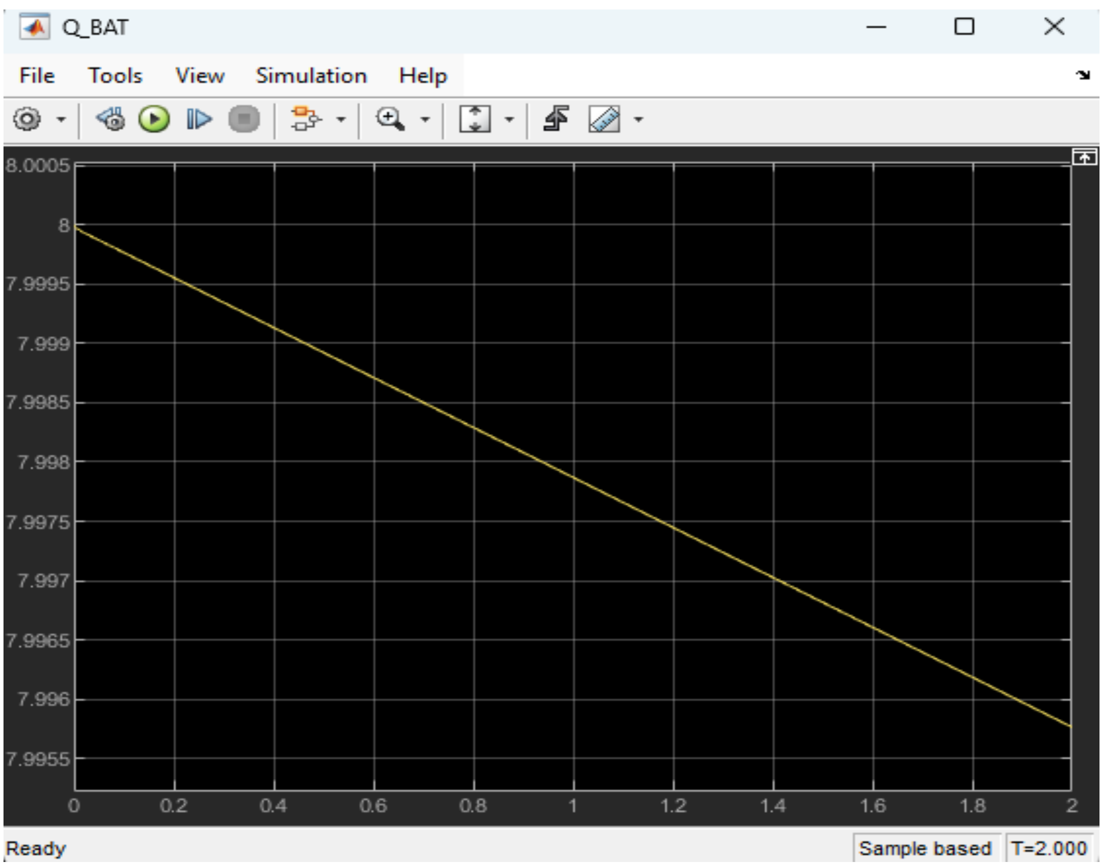
A local closed-loop controller was added to the battery subsystem to regulate the DC bus voltage. The controller measures the bus voltage V_bus and compares it with the reference value of 50 V. The difference between these two values is the voltage error. This error is multiplied by a small gain, then added to the base duty cycle of 0.6 to create a controlled duty-cycle command for the converter. A Saturation block was used to keep the duty cycle within a safe range, and a Unit Delay block was added so the converter uses the previous duty-cycle value while the controller calculates the next one. This avoids an algebraic loop and makes the controller behave more like a digital controller.

Test Objective and Results

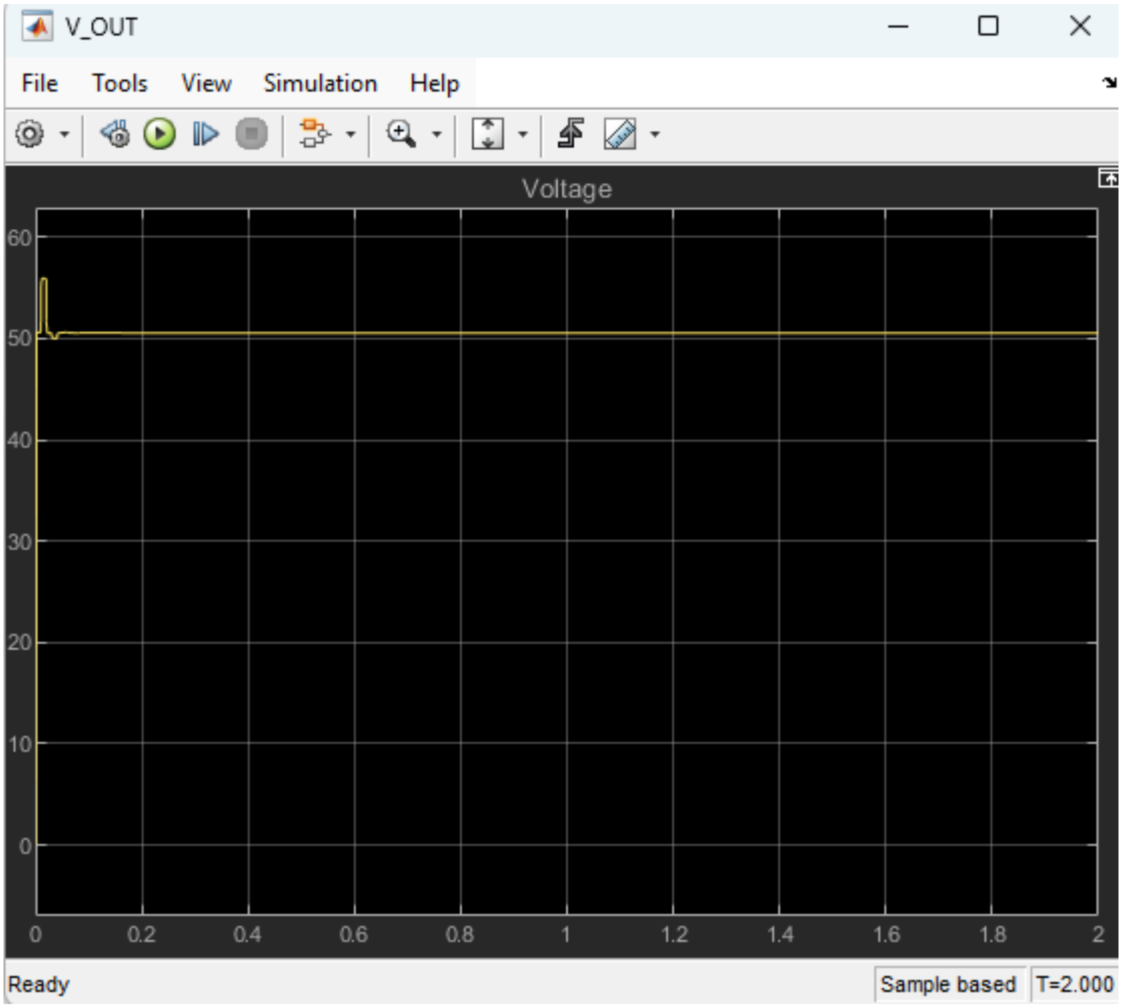
The goal of this test was to check whether the battery subsystem could supply the load and support the DC bus near 50 V. At first, the controller response was unstable, so the gain and duty-cycle limits were adjusted. After tuning, the subsystem became stable. The final test showed that the battery was discharging and supplying power to the load. The battery voltage was about 33.74 V, battery current was about 7.57 A, and battery power was about 255.5 W. On the bus side, the output voltage settled around 50.55 V, output current was about 5.05 A, and output power was about 255.5 W. This confirmed that the converter was stepping up the battery voltage while maintaining power balance.

Scope Observations

The Q_bat scope showed that the battery charge slowly decreased from about 8 Ah, confirming that the battery was discharging. The V_out scope showed a small startup transient and then settled close to the 50 V reference. The duty-cycle scope was used to monitor the actual duty cycle after the Unit Delay block, showing the controller command being sent to the converter. Overall, the test confirmed that the battery subsystem can support the DC bus and regulate the voltage close to the target value.



Q_bat or the scope showing battery charge.

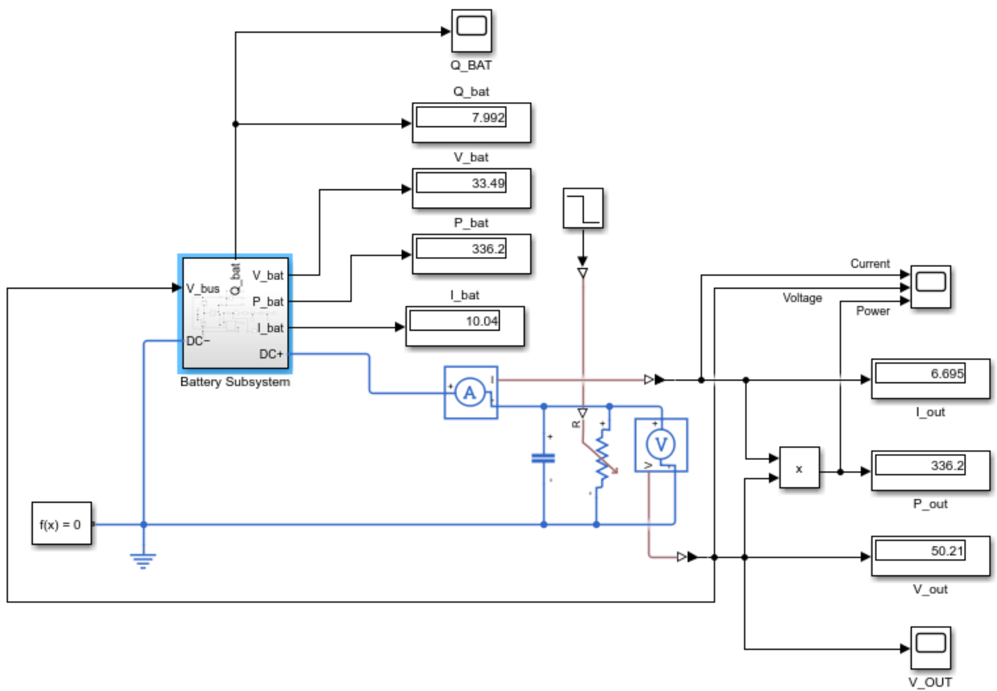


Bus Voltage

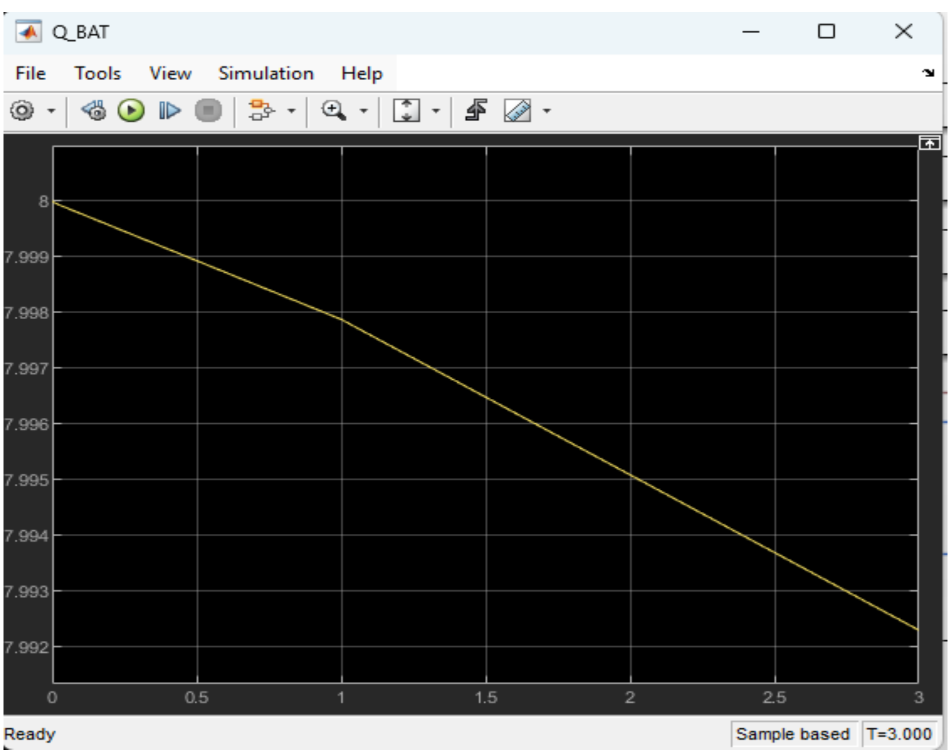
Battery Subsystem Load-Change Test

In this test, the fixed resistive load was replaced with a variable resistor to check how the battery subsystem responds when load demand changes. The load resistance was changed from 10 Ω to 7.5 Ω at t = 1 s, which increased the required load current and power. The local battery controller responded by slightly increasing the converter duty cycle, allowing the battery to supply more power to the DC bus. The bus voltage showed only a small adjustment at the load-change moment and then remained close to the 50 V reference. After the load change, the output voltage was approximately 50.21 V, the output current increased to about 6.695 A, and the output power increased to about 336.2 W. On the battery side, the battery supplied about 10.04 A and 336.2 W, while Q_bat decreased slightly faster after 1 s, confirming that the battery was discharging more quickly to support the higher load demand. This test confirmed that the battery subsystem controller can respond to a load increase and maintain the DC bus voltage close to the target value.

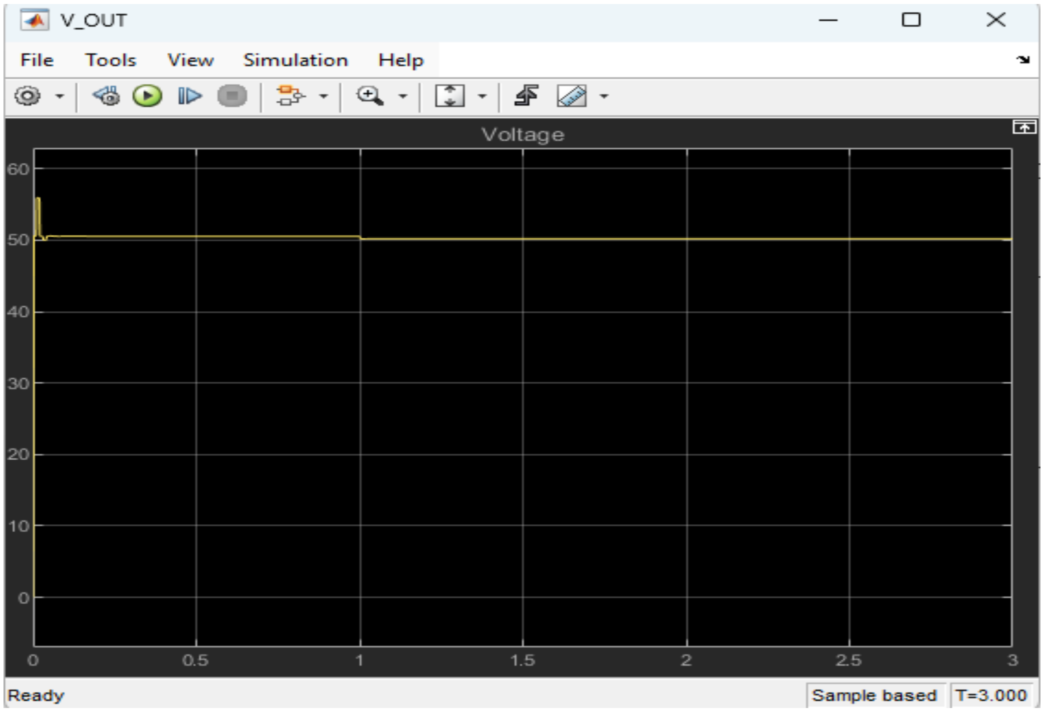
Battery subsystem connected to a variable load for load-change testing.



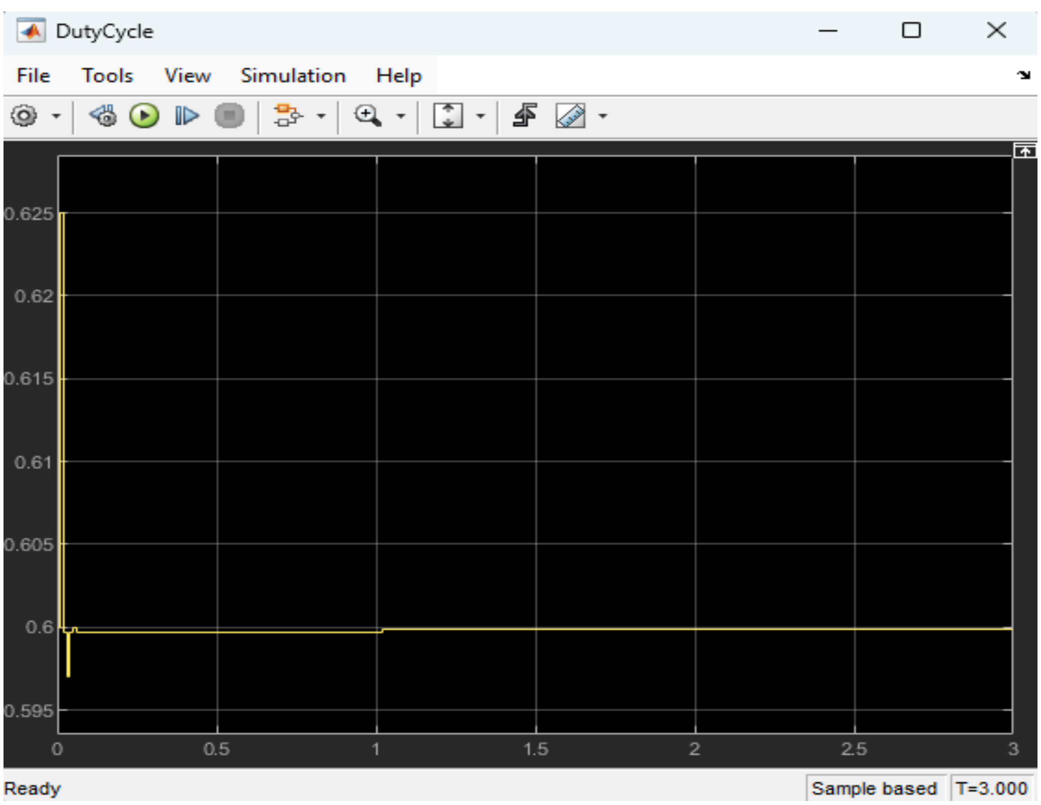
Battery charge Q_bat decreases faster after the load demand increases at 1 s.



DC bus voltage remains close to 50 V after the load-change disturbance.



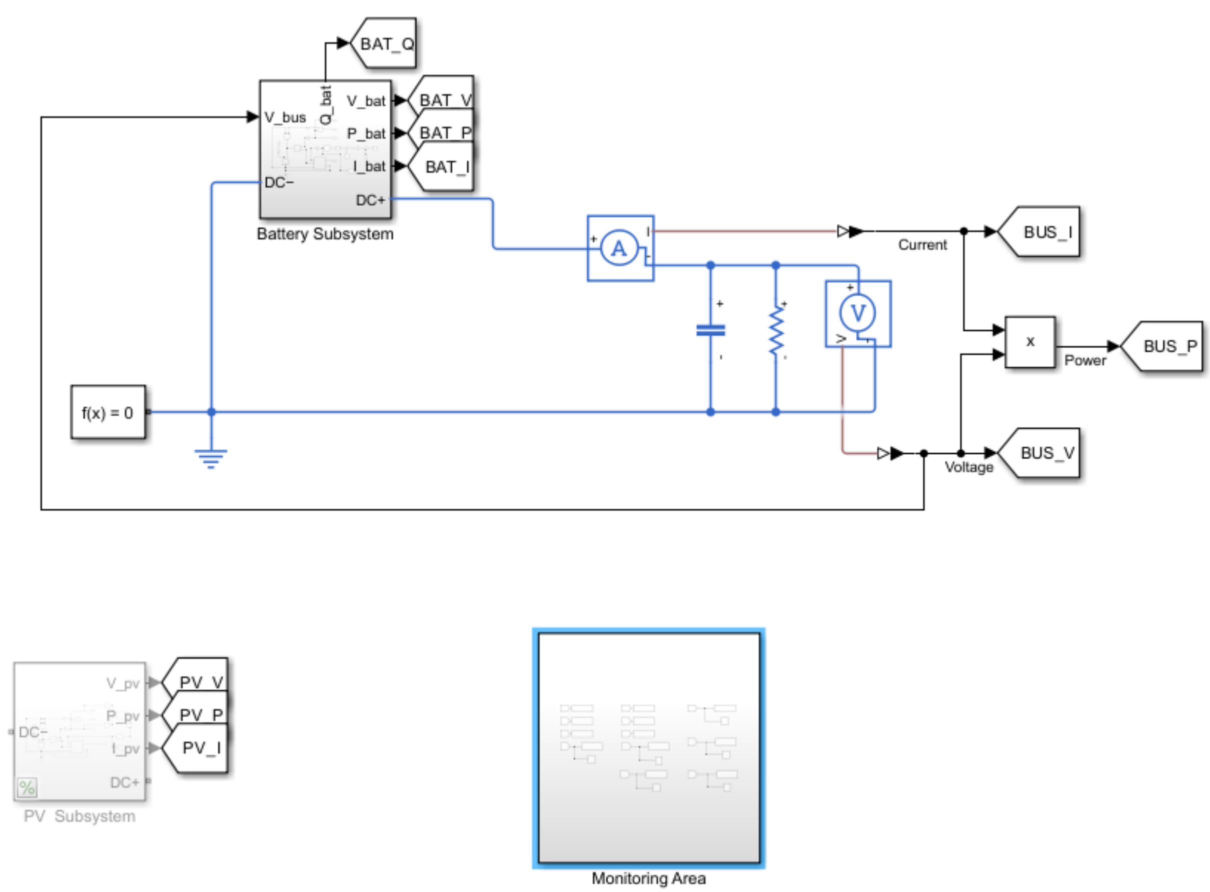
Duty cycle slightly increases after 1 s as the controller supplies more power to the bus.



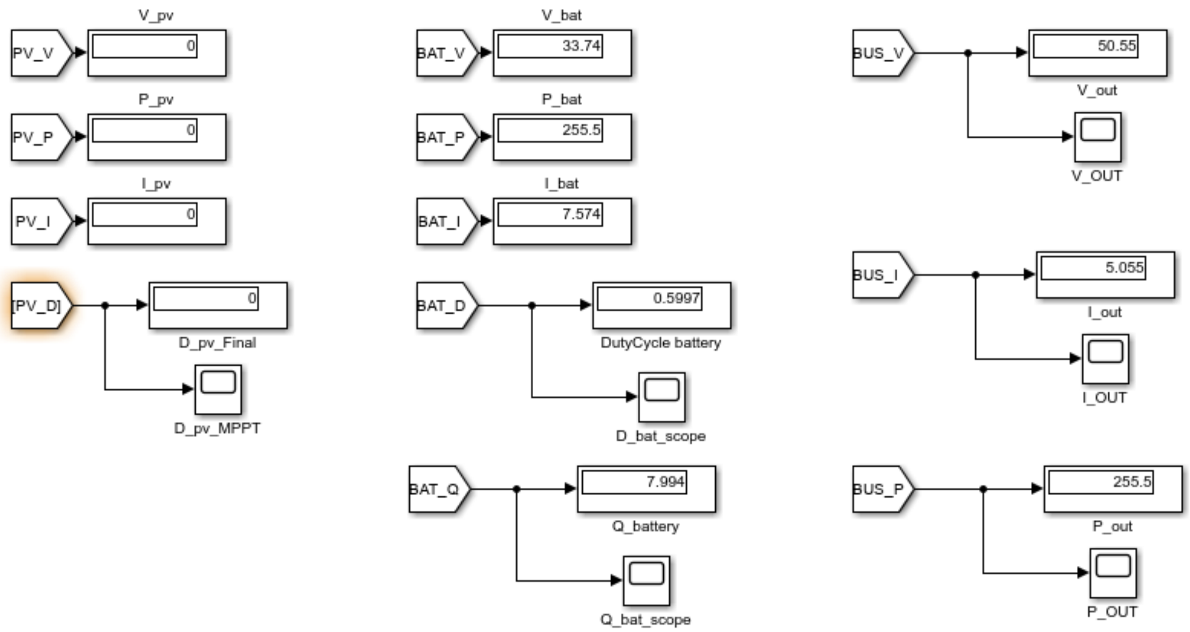
Model Cleanup Using Goto/From Blocks

As the model became larger, the number of measurement wires, displays, and scopes started making the main schematic difficult to read. To clean up the model, Goto and From blocks were used to route Simulink measurement signals without drawing long wires across the system. Each important signal was assigned a clear tag name, such as BAT_V, BAT_I, BAT_P, BAT_Q, BAT_D, BUS_V, BUS_I, and BUS_P. The Goto blocks were placed near the source of each signal, and the matching From blocks were placed inside a separate monitoring area/subsystem. This allowed all displays and scopes to be grouped in one location while keeping the main model focused on the physical power connections between the battery subsystem, DC bus, capacitor, and load. This cleanup makes the model easier to understand, debug, and expand when the PV and battery subsystems are integrated together.

Whole model



Monitoring Area/Subsystem

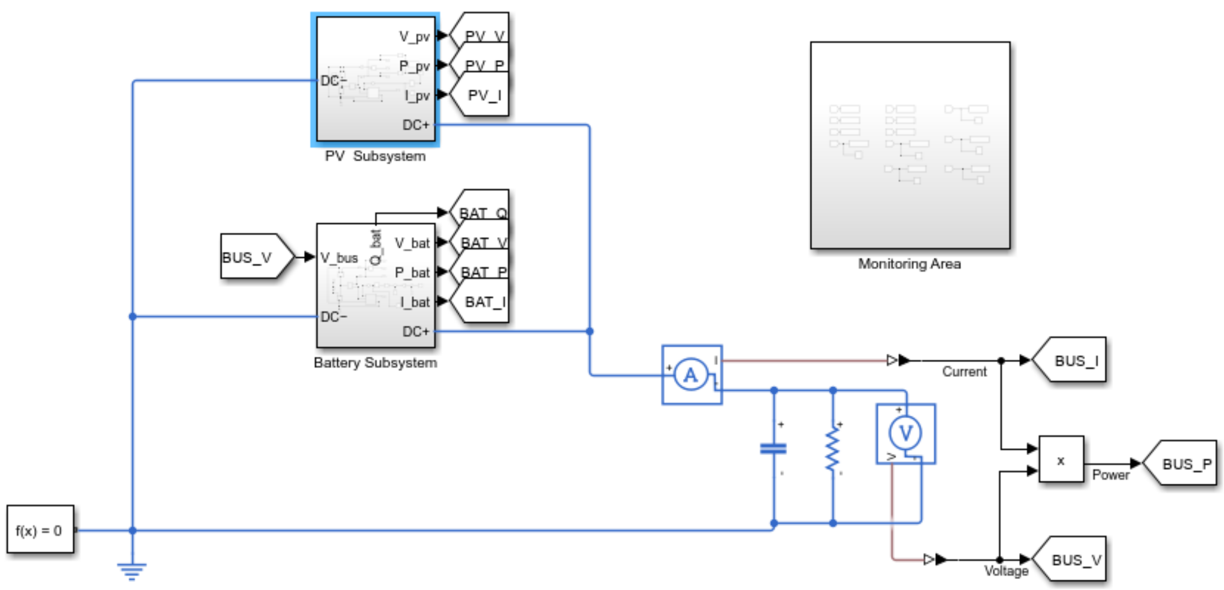


Battery Subsystem and PV Subsystem Integration

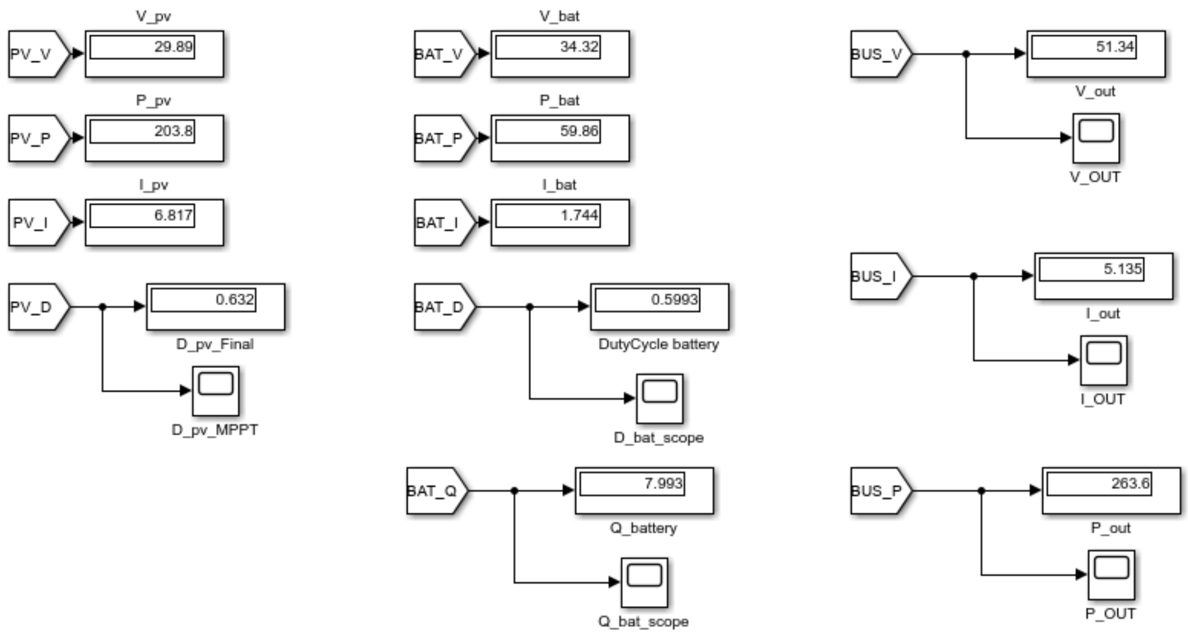
PV and Battery Integration Test

In this test, the PV subsystem and battery subsystem were connected in parallel to the same DC bus. The positive terminals of both subsystems were connected to the common DC bus positive line, and the negative terminals were connected to the common negative/reference line. The PV subsystem used its local MPPT controller to extract maximum available power from the PV source, while the battery subsystem used its local bus-voltage controller to support the DC bus voltage near the 50 V reference. After running the simulation, the PV subsystem produced about 203.8 W, while the battery supplied about 59.86 W. Together, they supplied the load power of about 263.6 W, which confirms that the power sharing was working correctly. The DC bus voltage stayed close to the target value at about 51.34 V, showing that the battery controller was helping regulate the bus while the PV subsystem supplied most of the load power. This confirmed that the PV and battery subsystems were successfully integrated on the same DC bus.

Integrated Model Schematic



Results



V_pv
PV_V
29.89
P_pv
PV_P
203.8
I_pv
PV_I
6.817
PV_D
0.632
D_pv_Final
D_pv_MPPT
V_bat
BAT_V
34.32
P_bat
BAT_P
59.86
I_bat
BAT_I
1.744
BAT_D
0.5993
DutyCycle battery
D_bat_scope
BAT_Q
7.993
Q_battery
Q_bat_scope
BUS_V
51.34
V_out
V_OUT
BUS_I
5.135
I_out
I_OUT
BUS_P
263.6
P_out
P_OUT